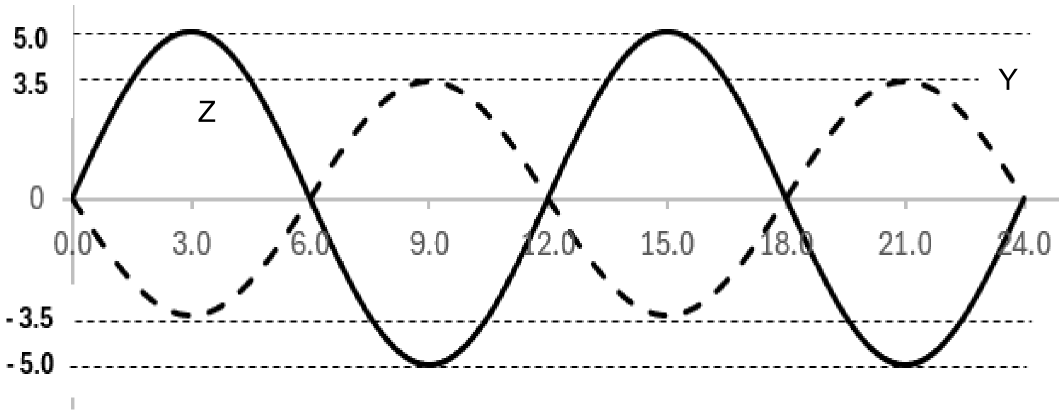
Q4	
(a)	Let the distance from the source to the first area be *r*. Using geometry, the distance from the point source is 7.5 times more for the second area. **Method I** $I \propto \frac{1}{r^2} \propto A^2$. Hence, amplitude is inversely proportional to the distance away from a point source, $A \propto \frac{1}{r}$ [M1] Since the distance from the point source is 7.5 times more for the second area, the amplitude would be 7.5 times less. Hence, the new amplitude would be $A_2 = \frac{A_0}{7.5} = 0.13A_0$ [A1] **Method II** Since the source is a point source, the intensity of the wave is inversely proportional to the square of the distance that the waves travel, $I \propto \frac{1}{r^2}$. Hence the intensity at the second area, I_2 $\frac{I}{I_2} \propto \frac{r_2^2}{r^2} = \frac{(7.5r)^2}{(r)^2}$ $I_2 = \frac{I}{7.5^2}$ [M1] Intensity is directly proportional to the square of the amplitude. Hence, the amplitude: $\frac{I}{I_2} \propto \frac{{A_0}^2}{A_2^2} \rightarrow A_2^2 = \frac{I_2}{I}{A_0}^2$ $A_2 = \frac{A_0}{7.5} = 0.13A_0$ [A1] **Method III** Intensity = Power / Area Intensity $\propto$ (1 / Area) $\propto$ (amplitude)^2 $\frac{I'}{I} = \frac{S}{S'} = \left(\frac{1.6}{12}\right)^2 \rightarrow I' = \left(\frac{1.6}{12}\right)^2 I$ [M1] $A' = \frac{1.6}{12}A_0 = 0.13A_0$ [A1]

(b)(i) 1 & 2	Wavelength = 1.2 m. **Two** full wavelengths of **stationary waves** must be drawn with • Nodes at 0.0 m, 0.6 m, 1.2 m, 1.8 m and 2.4 m. • Antinode peaks at 0.3 m and 1.5 m or at 0.9 m and 2.1 m [B1] **Maximum amplitude for Y at 12.5 ms** $y = 5.0\sin(\omega\, t) = 5.0\sin\left(\frac{2\pi}{T}t\right)$ $y = 5.0\sin\left(\frac{2\pi}{20}12.5\right) = -3.5$ mm [B1] **Maximum amplitude for Z at 5.0 ms** $y_Z = 5.0\sin(\omega\, t) = 5.0\sin\left(\frac{2\pi}{20}5.0\right)$ [B1] $y_Z = 5.0$ mm The antinode peaks for Z should coincide with antinode troughs for Y and vice versa [B1] 
(b)(ii)	180° or π rad